

I Thought I Knew SQL — Until I Saw These Query Patterns in Real Companies

SQL Query Patterns You'll See in Real Companies

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When I was learning SQL, most tutorials made it feel like a checklist of commands — `SELECT`, `JOIN`, `GROUP BY`, done. That illusion broke the moment I started working with real company data. Production SQL doesn't look clean, short, or academic. It follows patterns shaped by business rules, messy data, and constant change.

Over time, I noticed the same styles of queries repeating across projects — finance reports, dashboards, ingestion pipelines, and ad-hoc analysis. Below are the SQL patterns I see most often in real companies, explained the way they actually show up at work.



Working with dates the business way

In companies, time is rarely just a timestamp. Reports revolve around months, quarters, financial years, or billing cycles. That's why date logic appears in almost every serious query.

```
SELECT *  
FROM sales  
WHERE order_date >= DATE_TRUNC('month', CURRENT_DATE)  
      AND order_date < CURRENT_DATE + INTERVAL '1 day';
```

This kind of filtering avoids edge cases and matches how stakeholders think. It's also why you'll see developers avoid `BETWEEN` when timestamps are involved—it quietly creates bugs.

Aggregation with intent, not just GROUP BY

Aggregations in real systems are never neutral. They almost always include business rules: valid statuses, exclusions, or adjustments.

```
SELECT customer_id,  
       SUM(amount) AS total_spent  
FROM orders  
WHERE status = 'COMPLETED'  
GROUP BY customer_id;
```

The filter is not optional here. Without it, the number may be technically correct but practically useless.

Deduplication using window functions

APIs resend data. Jobs fail and rerun. Duplicate rows are normal. Companies handle this deterministically using window functions.

```
WITH ranked_data AS (  
  SELECT *,  
         ROW_NUMBER() OVER (  
           PARTITION BY order_id  
           ORDER BY updated_at DESC  
         ) AS rn
```

```
FROM orders_raw
)
SELECT *
FROM ranked_data
WHERE rn = 1;
```

This pattern keeps the latest version of each record and makes reruns safe.

JOINS that don't accidentally drop data

One of the most common production bugs comes from placing filters in the wrong place during a join.

```
SELECT o.order_id,
       c.customer_name
FROM orders o
LEFT JOIN customers c
  ON o.customer_id = c.customer_id
  AND c.is_active = TRUE;
```

By moving the filter into the join condition, teams protect the main dataset from disappearing rows.

Treating NULLs as a business problem

NULLs aren't technical quirks — they change financial results. That's why production queries handle them explicitly.

```
SELECT invoice_id,  
       COALESCE(adjusted_amount, 0) AS adjusted_amount  
FROM invoices;
```

Finance teams expect numbers, not blanks.

Encoding rules with CASE statements

When logic can't live in application code, it ends up in SQL.

```
SELECT amount,  
       CASE  
         WHEN amount > 10000 THEN 'HIGH'  
         WHEN amount BETWEEN 5000 AND 10000 THEN 'MID'  
         ELSE 'LOW'  
       END AS spend_category  
FROM transactions;
```

Risk bands, aging buckets, and classifications are often calculated this way.

Aging logic for invoices and vendors

A classic accounting requirement is to understand what's overdue and what isn't.

```
SELECT vendor_name,  
       CASE
```

```
        WHEN due_date < CURRENT_DATE THEN 'Overdue'  
        ELSE 'Not Due'  
    END AS payment_status  
FROM invoices;
```

In real dashboards, this expands into multiple time buckets feeding charts and alerts.

EXISTS for efficient checks

Sometimes you don't need data — you just need to know whether it exists.

```
SELECT customer_id  
FROM customers c  
WHERE EXISTS (  
    SELECT 1  
    FROM orders o  
    WHERE o.customer_id = c.customer_id  
);
```

This avoids heavy joins and scales better on large tables.

CTEs to make SQL readable for humans

Complex queries are written once and read many times. That's why teams rely on CTEs.

```
WITH monthly_sales AS (  
    SELECT DATE_TRUNC('month', order_date) AS month,  
           SUM(amount) AS revenue  
    FROM orders  
    GROUP BY 1  
)  
SELECT *  
FROM monthly_sales  
WHERE revenue > 100000;
```

CTEs make debugging and collaboration easier.

Final queries built for dashboards

BI tools prefer flat, predictable outputs. SQL does the transformation work upfront.

```
SELECT date,  
       SUM(revenue) AS total_revenue,  
       COUNT(DISTINCT customer_id) AS customers  
FROM sales_fact  
GROUP BY date  
ORDER BY date;
```

This pattern appears behind almost every production dashboard.

Closing thoughts

Real-world SQL isn't about clever syntax. It's about writing queries that survive bad data, changing rules, and real business pressure. Once you start recognizing these patterns, production SQL becomes easier to read — and easier to trust.

If you're learning SQL with the goal of working in real companies, focus on patterns like these. They matter far more than memorizing functions.